Mohammed Bilal P S

Ph.D. Researcher, Quantum Control Theory · Université de Bourgogne, France
Email · LinkedIn · Website

Research Overview

I am a second-year PhD researcher in quantum control theory at ICB, Dijon. My research spans analytical control theory, gradient-based numerical optimisation, and machine learning, using the Pontryagin maximum principle to formulate optimal control protocols for qubits subject to physical constraints (bounded control amplitude, energy cost, and robustness conditions), with GRAPE for numerical refinement.

I am now looking to engage with the hardware-aware side of this work: how protocols developed within a theoretical framework are evaluated, modelled, and adapted when the system is a real physical device with its own noise structure, timing constraints, and decoherence channels. Such a collaboration would clarify where the theoretical formulation meets experimental reality and where it breaks down.

Research Interests

- **Optimal and robust quantum control:** protocol design under physical constraints using the Pontryagin maximum principle and GRAPE; applications include energy-efficient state transfer and robustness against Hamiltonian uncertainty distributions
- **Open quantum systems and dissipative dynamics:** extending optimal and robust control to Lindblad master equations; how relaxation and decoherence channels constrain achievable protocols (active PhD direction)
- **Reinforcement learning and data-driven control:** learning and recovering optimal protocols from system data or partial model descriptions; reinforcement learning (Soft Actor-Critic) as a framework for protocol design when the Hamiltonian is not fully known

Education

Université de Bourgogne, Dijon, France *2024–2027 (ongoing)*
Ph.D. in Physics, Quantum Control Theory, ICB Laboratory (CNRS UMR 6303)
Supervisors: Dr. C. L. Latune, Prof. D. Sugny
Topic: *Optimal Quantum Control: Analytical and Numerical Approaches*

IISER Berhampur, India *2019–2024*
BS–MS Dual Degree in Physics, CPI 7.9/10
Supervisors: Dr. V. Mukherjee, Dr. P. A. Erdman
Thesis: *Reinforcement Learning for Robust Control of a Multi-Qubit Spin-Chain Quantum Battery*
Awarded the **Chanakya Fellowship** (iHub Quantum Technology Foundation, IISER Pune)

Research Projects

- **Universal Robust Quantum Control (Ph.D., ongoing):** Developing control protocols robust across Hamiltonian uncertainty distributions, combining Pontryagin-based analytical design with GRAPE refinement targeting 2nd and 4th order robustness conditions.
- **QOSTE: Energy Shortcut via Optimal Control (Ph.D., preprint 2025):** Proposed the QOSTE framework achieving STA-level state transfer at reduced energy cost; formulated as a constrained optimal control problem solved with GRAPE and given a geometric interpretation in the rotating frame. Submitted to PRL; arXiv:2503.20130.
- **RL for Spin-Chain Quantum Battery (MS thesis, 2024):** Applied Soft Actor-Critic to learn optimal charging protocols for spin-chain quantum batteries (2 to 5 qubits), extended to disordered systems and benchmarked against greedy strategies. Implemented in PyTorch.

Publications

- C. L. Latune, **M. B. P. Shebeek**, D. Sugny, *Energy Shortcut of N-Level Quantum Protocols by Optimal Control*, 2025. arXiv:2503.20130. Submitted to PRL.

Presentations & Posters

- **QTCC 2026, Cargèse, Corsica:** Poster, *Universal Robust Quantum Control of a Qubit*.
- **GdR TeQ^3 Colloquium, Grenoble (Nov 2025):** Poster, *Robust Optimal Control for a Two-Level System Using GRAPE*.
- **JED 2025, Dijon:** Poster, *Robust and Energy-Efficient Controls for a Two-Level System* (arXiv:2503.20130).
- **DITEQ Group Seminar:** Talk, *Steering Quantum Systems through Rewards: Reinforcement Learning Approaches*.

Technical Skills

- **Languages:** Python, C, C++, Bash
- **Quantum simulation:** QuTiP, NumPy, SciPy (time evolution, open system dynamics, control solvers)
- **Machine learning and RL:** PyTorch, Scikit-learn, policy gradient and actor-critic methods, PINNs